

C110 Programming Fundamentals I – Lessons 24/25

Campus Entry System



You are part of the campus safety team tasked with designing a digital solution to manage student entry during a pandemic. The system must be efficient, prevent unauthorised access, and support real-time reporting. Your goal is to develop a Python program that simulates a smart system to track student entry and exit, validate authorisation, and generate useful reports. The proposed system includes features such as recording student entry and exit, managing authorised users, tracking campus occupancy, and generating reports on unauthorised access attempts.

Requirements

Your program should include the following menu options:

- Record a student entering the campus
- Record a student leaving the campus
- Print all students currently on campus
- Add a student to the list of students approved to be on campus
- Print all unauthorised entry attempts

Additional requirements:

- Use functions to organise your program. Each menu option should call a separate function.
- Use lists to store authorised students, students currently on campus, and unauthorised attempts.
- Include validation to prevent double entry or exit when not on campus.

- Use loops and decision-making constructs to manage user interaction.

Reflection

- As a team, discuss the lessons learned from both the planning and implementation stages, and suggest how you would improve the next release of the system. This reflection should help you evaluate not only your programming solution, but also the effectiveness of your design and prioritisation decisions.

A SAMPLE PROGRAM OUTPUT on the next page

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Enter student ID to authorise:

S1001 added to authorised list.

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Enter student ID to authorise:

S1002 added to authorised list.

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Enter student ID:

Entry recorded for S1001

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Enter student ID:

Exit recorded for S1001

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Enter student ID: S1002

Error: S1002 is not on campus.

1. Record a student entering the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Enter student ID:

Unauthorised entry attempt by S1234

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Unauthorised attempts:

S1234

1. Record a student entering the campus
2. Record a student leaving the campus
3. Print all students currently on campus
4. Add a student to the authorised list
5. Print unauthorised entry attempts
6. Exit

Enter option (1-6):

Existing system

Continue on next column